

2019 E-Commerce Sales Analytics Dashboard

1. Project Overview

This Tableau project analyzes 2019 e-commerce sales data from 12 monthly CSV files. The goal is to understand sales trends, top products, city-level performance, product demand, and hourly purchase behavior.

- Dashboard 1: Executive Sales Overview
- Dashboard 2: Product & Customer Insights

2. Data Preparation

The 12 monthly sales files were combined using a Wildcard Union because all files had the same structure.

Union pattern used: Sales_*.csv

Basic cleaning filters were applied to remove invalid rows:

- Order ID is not null
- Product is not null
- Product is not equal to "Product"
- Quantity Ordered > 0
- Price Each > 0

3. Key Calculations

Calculation	Formula	Purpose
Revenue	[Quantity Ordered] * [Price Each]	Calculates total revenue for each order line.
Average Order Value	SUM([Revenue]) / COUNTD([Order ID])	Shows average revenue generated per order.
Month Year	DATETRUNC('month', [Order Date])	Groups sales by month for trend analysis.
Hour of Day	DATEPART('hour', [Order Date])	Analyzes hourly purchasing behavior.
City	TRIM(SPLIT([Purchase Address], ', ', 2))	Extracts city from purchase address.
State	TRIM(LEFT(TRIM(SPLIT([Purchase Address], ', ', 3)), 2))	Extracts state abbreviation from address.
Sales Contribution %	SUM([Revenue]) / TOTAL(SUM([Revenue]))	Shows product share of total revenue.

4. Top N Products Parameter

A Top N Products parameter allows users to dynamically view Top 5, Top 10, Top 15, or Top 20 products based on revenue.

5. Dashboard 1: Executive Sales Overview

This dashboard provides a high-level view of overall sales performance.

Includes

- Total Revenue
- Total Orders
- Quantity Sold
- Average Order Value
- Monthly Revenue Trend
- Top N Products by Revenue
- Revenue by City Map

Business Questions Answered

- What was the total sales performance?
- Which months generated the most revenue?
- Which products drove the most revenue?
- Which cities had the strongest sales?

6. Dashboard 2: Product & Customer Insights

This dashboard provides deeper analysis of products, cities, and purchase behavior.

Includes

- Quantity Sold by Product
- Revenue vs Quantity Scatter Plot
- Top Cities by Revenue
- Product Revenue Contribution %
- Revenue by Hour of Day

Business Questions Answered

- Which products sold the most units?
- Which products generated the highest revenue?
- Which cities contributed the most sales?
- What time of day generated the highest revenue?
- Which products contributed most to total revenue?

7. Filters and Navigation

Both dashboards use shared filters so users can narrow the analysis by time, product, and geography.

- Month Year
- Product
- City
- State
- Top N Products

Navigation buttons connect both dashboards:

- Dashboard 1 to Dashboard 2: Product & Customer Insights ->
- Dashboard 2 to Dashboard 1: <- Executive Overview

8. Tableau Skills Demonstrated

• Wildcard Union	• Scatter plot
• Data cleaning filters	• Dynamic Top N parameter
• Calculated fields	• Product ranking
• KPI cards	• Percent-of-total calculation
• Line chart	• Shared filters across dashboards
• Bar charts	• Dashboard navigation buttons
• Map visualization	• Interactive business storytelling

9. Project Summary

This two-dashboard Tableau workbook analyzes 2019 e-commerce sales performance. It provides an executive overview of revenue, orders, products, and cities, along with a deeper product and customer insights page. The dashboard helps users explore sales trends, product demand, geographic performance, and hourly purchase behavior.